Bachelor Thesis Proposal

Frequency Changes in Late Spring Frost Events after the Start of Season and Their Impacts on Carbon Fluxes in Forest Ecosystems

Summary

Climate warming has advanced spring phenology in many temperate and boreal forests, but earlier start of season (SOS) may also increase exposure to damaging frost events. Such late spring frost (LSF) events may reduce ecosystem carbon uptake, yet their occurrence after SOS and their effects on forest carbon fluxes remain insufficiently quantified across sites and years. This thesis therefore investigates how frequently frost occurs after SOS at forest eddy-covariance sites and how such events affect ecosystem carbon fluxes, with a particular focus on gross primary production (GPP). The analysis will use harmonized FluxDataKit data, including eddy-covariance flux measurements and site-level meteorological variables. Late spring frost events will be identified using both temperature-based and flux-based definitions in order to compare alternative approaches to event detection. Event frequency, timing, and spatial patterns will be analysed across sites and years, and short-term GPP reductions following frost events will be quantified. The project will improve understanding of how changing spring phenology and frost risk interact to affect forest carbon uptake.

Background

Climate warming has advanced spring phenology across many temperate and boreal forests, but earlier spring development does not necessarily translate into higher ecosystem carbon uptake. Instead, an earlier start of season (SOS) can increase the exposure of vegetation to freezing temperatures during sensitive developmental stages (Liu et al., 2018; Chamberlain et al., 2021). At broader spatial scales, this risk shows strong regional variation, indicating that warming-driven phenological shifts and frost exposure do not change uniformly across regions (Zohner et al., 2020). In Europe, especially, earlier leaf-out in deciduous trees has been linked to increasing false spring risk, particularly for early-leafing species, whereas later-leafing species appear more stable (Chamberlain et al., 2021).
A central challenge in this field is how late-spring frost events should be defined. Chamberlain et al. (2019) argue that fixed calendar windows and universal temperature thresholds are often insufficient, because frost risk depends on species, phenological stage, and regional climate conditions. They recommend biologically meaningful risk periods linked to spring development. This is also supported by studies showing that accumulated spring warmth, such as growing degree days (GDD), is important for assessing how vulnerable vegetation is when frost occurs (Zohner et al., 2020). At the same time, warmer winters may reduce chilling accumulation, delay leaf unfolding, and prolong the period of vegetative risk, especially when late frost occurs repeatedly (Chamberlain & Wolkovich, 2021). Repeated frost events can further reduce resilience and affect growth, although species differ strongly in their responses (Hufkens et al., 2012; Rubio-Cuadrado et al., 2021). Frost impacts may also interact with drought, adding further complexity to forest responses (Charrier et al., 2021).

Beyond phenology, late-spring frost can affect ecosystem carbon uptake. Eddy-covariance studies have shown that frost events reduce gross primary production (GPP) and cumulative CO_2 uptake at forest sites, with impacts ranging from immediate declines to carry-over effects into the following year (Hufkens et al., 2012; Chen et al., 2023; Yan et al., 2023). These effects are important enough that neglecting frost in ecosystem models can lead to overestimated forest productivity (Liu et al., 2018; Meyer et al., 2024).
Against this background, this thesis investigates how frequently frost occurs after SOS at forest eddy-covariance sites and how such events affect ecosystem carbon fluxes, with a particular focus on GPP. By comparing temperature-based and flux-based event definitions, this thesis aims to assess the ecosystem carbon responses to late spring frost events.

Objectives and research questions

Primary objective

This project aims to quantify the occurrence of frost after SOS and evaluate its short-term impact on carbon fluxes using eddy-covariance measurements.

Methodological basis

1. How can late frost events be consistently identified at eddy-covariance forest sites?

2. How do event definitions based on thermal forcing (GDD) compare with flux-based definitions using seasonal GPP development?

Core analysis

3. How frequently do late frost events occur after SOS, and how do their frequency and timing vary across sites over time?

4. What is the magnitude and duration of GPP reductions following late frost events?

Additional analysis

5. Do late frost events also affect vegetation greenness indicators such as the NDVI?

6. Does the impact of late frost events depend on the timing of the frost event relative to SOS?

Data and study system

The analysis will use FluxDataKit (GECO Bern), which provides harmonized eddy-covariance flux data and site-level meteorological drivers such as air temperature: https://geco-bern.github.io/FluxDataKit/index.html

These data enable identifying frost events directly from measured temperatures and quantifying carbon-flux responses at daily to weekly timescales.

Methodological approach and implementation

1. Identification of late spring frost (LSF) events

First, late spring frost events will be identified for each site-year. This will be done in two steps: (1) using a temperature-based definition linked to spring development, for example based on growing degree days (GDD) (Zohner et al., 2020), and (2) using a flux-based definition in which frost events are only counted after ecosystem productivity has already increased beyond an early-season threshold, for example after daily GPP exceeds 10% of the winter-to-summer seasonal amplitude (Chen et al., 2023).

2. Temporal and spatial patterns of LSF occurrence

For each site-year, the number of late frost events, the number of frost days after SOS, and the timing of the last frost relative to SOS will be calculated. These metrics will then be compared across sites, years, and forest types in order to assess temporal trends and spatial differences.

3. Impacts of LSF on Gross Primary Production (GPP)

The short-term impact of identified late frost events on gross primary productivity will be quantified using event-based analyses. GPP anomalies in the days following each event will be compared to pre-event conditions and to an expected seasonal baseline. Where possible, meteorological covariates such as radiation and temperature will be considered to separate frost effects from background variability.

4. Timing sensitivity and ecosystem differences

A further analysis will test whether the GPP response depends on the timing of the event relative to SOS and whether responses differ systematically among forest types.

5. Vegetation greenness indicators

If suitable remote-sensing data are available, the analysis will be extended to vegetation greenness indicators such as the NDVI to test whether late frost impacts are also visible in canopy greenness.

Risk and contingency

A key methodological risk of this project is the lack of a universally accepted definition of late frost events after start of season (SOS). Different event definitions may identify different frost occurrences and thus affect estimates of both event frequency and ecosystem response. To address this, the study will compare two complementary approaches: a temperature-based definition linked to spring development and a flux-based definition based on seasonal GPP dynamics. In addition, sensitivity analyses with alternative thresholds will be used to test the robustness of the results.
Another challenge is that some sites or years may contain only few clear late frost events after SOS, which may limit the statistical power of event-based analyses. To mitigate this, this thesis will include multiple forest sites and years. That way the number of events will hopefully be sufficient to be able to conduct thorough analysis.

Literature

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